

DTHFormer: Dual Token Hierarchical Transformer Network for Fine-Grained Emotion Recognition

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Abstract

The recognition of emotions through physiological signals has consistently remained a popular area of research. While utilizing physiological signals for emotion recognition provides the advantages of authenticity and objectivity, there remains room for improvement, particularly in subject-independent experimental conditions, due to the significant variability in physiological responses among different individuals. The development of multi-modal interaction methods offers the possibility of aiding emotion recognition based on physiological signals by introducing cross-domain information. This paper proposes a Dual Token Hierarchical Transformer Network to recognize fine-grained valence and arousal labels, which is dedicated to effectively integrating physiological data and emotional stimulus data. Specifically, DTHFormer performs indirect interaction through dual token between physiological signal and visual/audio modal extracted from video in order to protect unique information of each modality and avoid information confusion. Then, a hierarchical fusion module is used to preserve key emotional information across multiple modalities, enabling the model to have a deeper comprehension of emotional information. A variety of classification experiments demonstrate that our proposed method achieved good results. Further ablation experiments show the effectiveness of our proposed modules.

Index Terms- *fine-grained emotion recognition; dual token; hierarchical fusion; emotional stimulus; physiological signals*

I. INTRODUCTION

WITH the rapid development of social media platforms and artificial intelligence, people's emotions are increasingly susceptible to the influence of digital information. Meanwhile, various fields such as mental health assessment and detection, personalized human-computer interaction, and immersive gaming experience require recognizing users' emotions. Analyzing users' physiological signals, facial expressions, voice intonation, and similar cues to infer emotions has emerged as a prominent research area, i.e. emotion recognition.

Research in emotion recognition initially focused solely on physiological signals. Current approaches can be categorized into those utilizing single physiological signals and those employing multiple signals. For single signal methods, existing techniques include EEG-based methods [1], [2], [3], ECG-based methods [4], [5], [6], and fNIRS-based methods [7], [8], [9]. In contrast, the prevalent methods for multiple signal analysis [10], [11], [12] integrate various types of physiological signals to enhance emotion recognition as shown in Fig. 1(a).

While emotion recognition research based on physiological signals offers benefits in terms of authenticity and objectivity, it also faces several limitations. The presence of noise interfer-

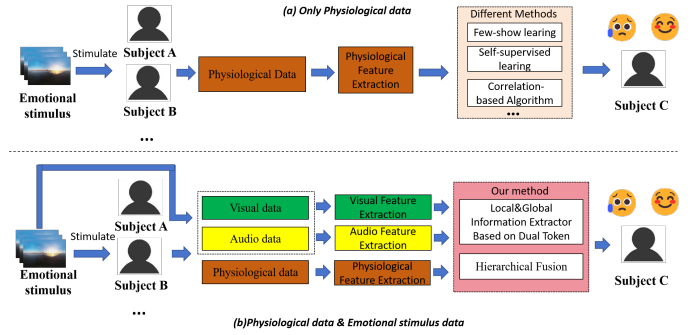


Fig. 1: Comparison of existing methods and our method in emotion recognition based on continuously self-annotated physiological signals. (a) Methods only using physiological data. (b) Our method using physiological data and emotional stimulus data.

ence, the stringent requirements for collection equipment, and particularly the significant inter-subject variability contribute to the ongoing challenge of accurately recognizing emotions from the physiological signals of unseen individuals. To address these complexities, previous research has carefully examined the nature of emotional responses, attempting to delineate the boundaries between individual reactions and common emotional patterns. James et al. [13] examined emotional responses to film stimuli in a large ethnically diverse sample of 494 subjects and identified sizeable individual variability in emotional responses to film stimuli, with notable differences in the discreteness and intensity of elicited emotions across individual subjects and target emotion categories. Boucher and Brant [14] found commonalities in emotion antecedents in the many non-Western cultures they examined. It was not in the specific details but at a more abstract level that universality in antecedent events was found. These research suggests that although emotional experiences exhibit significant individual differences, the stimuli that evoke emotions still share certain commonalities in the abstract space. What's more, Lauri et

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al. [15] found that negative valence synchronizes individuals' brain areas supporting emotional sensations and understanding of another's actions, whereas high arousal directs individuals' attention to similar features of the environment. Synthesizing these findings, although the physiological manifestations of emotions are highly individual-specific, stimuli can still provide a common and stable semantic reference in emotion recognition. In fact, many studies have integrated audio-visual features to estimate intrinsic emotions from video content [16], [17], [18]. Previous studies have shown that EEG signals are highly correlated with the emotional information involved in multimedia content (visual and audio) [19], and combining the two for emotion recognition can achieve better results [20]. In summary, video and audio modalities represent the objective properties of stimulus, which are constant for all subjects in the experiment. The common emotional information they contain can often serve as a reference, helping models mitigate individual differences in physiological signals. Specifically, by associating physiological responses to audio-visual context, the model establishes a common ground for cross-subject comparison. Therefore, for cross-subject emotion recognition, effectively encoding and integrating these two data modalities can extract pertinent emotional cues from videos and enhance both the performance and robustness of the model.

Inspired by the above situation, a Dual Token Hierarchical Transformer Network (DTHFormer) is proposed in this paper. As shown in Fig. 1(b), we use dual-token indirect interaction to preserve modality-specific representations and hierarchical fusion to progressively aggregate emotion information, in order to alleviate the instability of subject-independent emotion recognition. Specifically, we utilize the Local Information Extractor to extract emotional information contained in the video content and the Global Information Extractor to maintain global information consistency among the three modalities, thereby providing a stable reference to mitigate individual differences. Otherwise, direct fusion may lead to information confusion, which stems from the fact that distinct modalities encode emotion differently, despite each conveying emotionally relevant cues. As Juan [21] stated, the academic community has not reached a unified standard on how to define the emotional response patterns of various dimensions. That is, for physiological-level, similar physiological responses may be paired with different emotional labels. For instance, when watching a horror movie, one subject may perceive fear while another finds it thrilling, yet both of them exhibit increased heart-rate and skin-conductance. This inspires us to preserve modality-specific information before fusion, then perform hierarchical fusion to retain key emotional information. In general, the dual-token scheme keeps separate tokens for vision-audio and physiology modules, so each stream first learns its own pattern without mixing conflicting cues. Then, by preserving key information in multi-modal data through hierarchical fusion module, the model can more accurately understand emotional information and more accurately recognize emotion.

The main contributions can be summarized as follows:

- A novel architecture (DTHFormer) is proposed to conduct fine-grained emotion recognition based on dual token hierarchical transformer network, which integrates

physiological and emotional stimulus data to recognize emotion. By integrating the information from emotional stimulus and physiological data, our model is capable of more accurately recognizing emotions in unseen subjects on each fine-grained segment.

- A Multi-modal Interaction Module Based on Dual Token is proposed to extract information from both local and global dimensions. This module assists the model to extract information between emotional stimulus and physiological data, and learn the correlations between each modal.
- A Hierarchical Fusion Module is introduced to reduce information redundancy by leveraging cross-domain feature correlations to retain essential information. This module fuses the information between physiological modal and emotional stimulus modal in order to more accurately understand emotional information.
- Various types of experiments indicate that the proposed method outperforms existing approaches on three emotion datasets with continuous self-annotation and labels. Moreover, ablation experiments provide evidence of the efficacy of using indirect interaction methods based on dual token to introduce emotional stimulus information in fine-grained emotion recognition.

II. RELATED WORK

A. Emotion Recognition Based on Physiological Signals

Physiological signals used for emotion recognition mainly include electroencephalography (EEG), functional near-infrared spectroscopy (fNIRS), heart rate (HR), electrodermal activity (EDA), etc. Research based on physiological signals use hand-crafted features at first. Duan et al. [22] introduced a feature extraction technique known as differential entropy and performed emotion recognition experiments on the SEED dataset to validate the efficacy of the proposed feature. Soroush et al. [23] reconstruct EEG phase space and transform it into a new state space, which named Poincare planes, to extract features mathematically. Moon et al. [24] considered brain connectivity features in the emotion recognition task to effectively capture asymmetric brain activity patterns. Picard et al. [25] proposed six traditional statistical features commonly used for physiological signals, including the mean, standard deviation, mean of the absolute value of the first-order difference, mean of the absolute value of the second-order difference, and mean of the absolute value of the first-order and second-order differences of the normalized signal. Yang et al. [26] extracted features by performing discrete wavelet transform on Electromyography.

The advancement of deep learning shows the superiority of deep features compared to traditional manual features. Choi et al. [27] fused video modality and EEG modality and used low-rank decomposition and bilinear pooling to calculate continuous emotion. Nagasawa et al. [28] estimated the continuous emotion of subjects over time by measuring facial expressions and physiological signals including EEG and heart rate. Becker et al. [29] evaluated the emotion recognition accuracy using various combinations of physiological signals, including EEG, galvanic skin response, ECG, respiration, blood oxygen levels,

and pulse. Wang et al. [30] used short-time Fourier transform to extract electrode frequency distribution maps and used deep convolutional neural networks based on residual blocks for feature extraction and emotion classification. Du et al. [31] proposed an emotion recognition model based on long short-term memory network (LSTM) to characterize the nonlinear relationship between EEG signals of different electrodes. Li et al. [32] introduced a spatiotemporal graph neural network to capture the graph relationship representation among emotional EEG signals. Yang et al. [33] proposed a novel attention-based LSTM system which combines user's behavioral and physiological features to determine user's emotional states. Dong et al. [34] developed a Pre-trained model based Multimodal Mood Reader for cross-subject emotion recognition that utilizes masked brain signal modeling and interlinked spatial-temporal attention mechanism. Jia et al. [35] proposed MDNet which consist of a modality-level disentangling module and a subject-level disentangling module. The modality-level disentangling module projects multimodal physiological signals into modality-invariant subspace and modality-specific subspace, capturing modality-invariant features and modality-specific features.

Utilizing physiological signals for emotional recognition has emerged as a significant research direction. However, there are still problems especially the significant inter-subject variability. Due to the different physiological conditions of individuals, there is no one-to-one correspondence between emotions and physiological signals. While existing studies have attempted to address this through few-shot learning [12], self-supervised learning [10], masked brain signal modeling with spatial-temporal attention [34], multi-level disentangling network [35], these approaches often ignore global semantic information in the latent space. Moreover, most methods rely on EEG signal.

B. Emotion Recognition Based on Videos

In the field of emotion recognition, videos are usually used as emotional stimulus sources because of the abundance of information, which can induce more real and natural emotion. Zhang et al. [36] introduced an innovative multi-modal emotion recognition model for conversational videos, leveraging reinforcement learning and domain knowledge. They employed a deep-Q-network based on gated recurrent units (GRUs) to learn the appropriate actions. Gunawan et al. [37] illuminated the process of recognizing emotions from videos using deep learning techniques, offering a detailed methodology and description of the recognition process. Zhao et al. [38] propose an end-to-end approach base on convolutional neural networks(CNNs) and design a special classification loss to recognize video emotions. Middy et al. [39] explore model-level fusion to identify the optimal multi-modal model for emotion recognition, leveraging both video and audio modalities. Hazarika et al. [40] introduce a conversational memory network, which utilizes contextual information to identify utterance-level emotions in dyadic conversational videos.

Many methods have been dedicated to recognize emotion based on videos. However, most of these approaches focus

on studying the emotional information contained in the video itself. Meanwhile, modal interactions are mostly limited to visual modal, audio modal, and text modal.

C. Multi-modal Emotion Recognition

Multi-modal continuous emotion recognition is a technology integrating various information sources to recognize subjects' continuous emotion. Park et al. [41] proposed a multi-modal continuous emotion dataset, which contains physiological signals, audio, electroencephalogram and other data as well as the user's continuous emotion annotations during the conversation. The introduction of this dataset has promoted the research on emotion personalization in natural environments. Zhang et al. [42] used a unique attention mechanism (leader-follower attention) and combined it with the spatiotemporal coding structure to recognize continuous emotion based on the integration of linguistic information and audio-visual information. Huang et al. [43] introduced a model-level fusion approach that integrates Transformer and long short-term memory networks (LSTM) to combine audio and visual modalities, thereby enhancing the accuracy of multi-modal continuous emotion estimation. Li et al. [44] introduced an innovative multi-modal shared network with a cross-modal distribution constraint to address the issue of missing modalities. By integrating visual and physiological modalities, they enhance the robustness of the model. Raju et al. [45] explore the effectiveness of cutting-edge deep learning architectures that incorporate recurrent neural network (RNN) variants and explicit attention mechanisms. Their goal is to close the cultural gap (between German and Hungarian contexts) in predicting emotions from videos. Tsai et al. [46] Multimodal Transformer (MulT) for analyzing human multimodal language. MulT serves as a strong baseline capable of capturing long-range contingencies, regardless of the alignment assumption. Lin et al. [18] study the capability of frozen ViTs, pretrained only on visual data, to generalize to audio-visual data without finetuning any of its original parameters. Hou et al. [47] design a Semantic Alignment network based on Multi-Spatial learning (SAMS) for multi-modal emotion recognition, which achieves local and global alignment between modalities using high-level emotion representations of different modalities as supervisory signals. Ma et al. [48] propose SDT, a transformer-based model with self-distillation for multimodal emotion recognition in conversations. They use intra- and inter-modal transformers to model intra- and inter-modal interactions between conversation utterances. Salas-Cáceres et al. [49] conducted multimodal emotion recognition based on a fusion of audiovisual information with temporal dynamics. Kumar et al. [50] proposed a novel multi-modal interpretable depression detection system integrating visual, physiological, audio, and textual data.

Multi-modal emotion recognition provides a more comprehensive and robust affective representation than unimodal approaches. However, multi-modal models frequently face the challenge of information confusion, when processing data with large differences. Although existing methods capture the complementarity between different modalities through feature-level fusion, decision-level fusion, and cross-modal interac-

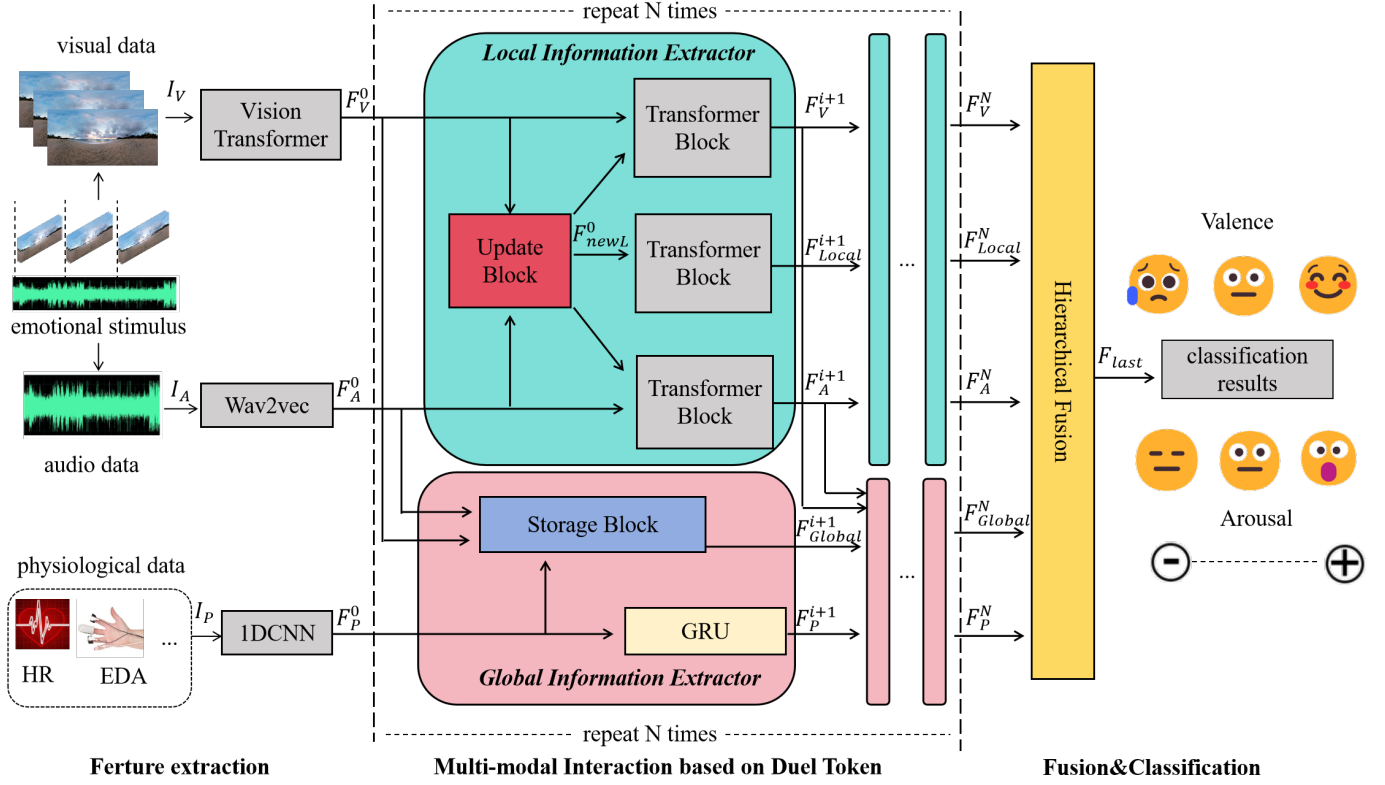


Fig. 2: Overall architecture of the proposed DTHFormer with three stages. At first, the raw data of three modalities are fed into their respective feature extractors to obtain multi-modal feature and initialize dual token. In the second stage, dual token is used to assist multi-modal interaction. Specifically, utilizing update block to update the information contained in local token and transfer it to visual and audio modal. At the same time, utilizing storage block to update the information contained in global token but do not transfer it to any modal. Finally, after the hierarchical fusion module, the fine-grained emotion is obtained on each segment.

tion, they often ignore global semantic alignment and fine-grained interactions.

III. PROPOSED METHOD

The proposed model, depicted in Fig. 2, is intended to first reduce cross-modal interference and then enhance emotion information aggregation through three stages. In the initial stage, after the data preprocessing, the multi-modal feature extractor based on pre-trained model is used for extracting features from visual, audio and physiological data. The initialization of dual token will also be carried out at this stage. In the second stage, visual features interact with audio features via local token, while both modalities interact indirectly with physiological features through global token to avoid mixing conflicting cues. In the final stage, a hierarchical module is used to fuse multi-modal features and preserve key information for robust prediction. Further illustration of these modules will be provided in the subsequent subsections.

Frame images and audio are separated from original videos to obtain the input data, i.e. frame image I_V and audio I_A . Visual features are extracted using vision transformer model [51], and audio features are extracted using wav2vec2 model [52]. For visual feature, the feature $F_V^0 \in R^{L \times D}$ is obtained through linear interpolation after vision transformer. For audio feature, the feature $F_A^0 \in R^{L \times D}$ is divided into segments of two seconds after the wav2vec2 model. During the entire process, the parameters of those two feature extraction

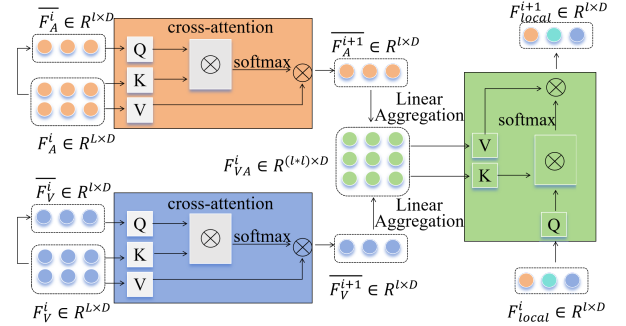


Fig. 3: An illustration of Update Block.

models are frozen and no gradient update and optimization are performed.

A. Data Preprocessing & Feature Extraction

This module is used to extract features for subsequent operations. Here 1DCNN is used for feature extraction following the previous work of Zhang et al. [53]. First, various physiological data collected by different sensors are resampled to unify their sampling rates. Then, the physiological signal $I_P \in R^{L_p \times D_p}$ is normalized to reduce the impact of individual differences from the data source, while D_p represents the number of selected physiological signal types, L_p represents the length of each physiological signal. In each type of physiological signal, the

i -th signal is recorded as S_{ij} , and the maximum and minimum values in the sequence are recorded as S_{maxj} and S_{minj} , which are expressed as follows:

$$I_P = [S_{ij}; \forall i = 1, \dots, L_P; \forall j = 1, \dots, D_P] \quad (1)$$

$$Normalize(S_{ij}) = [\frac{S_{ij} - S_{minj}}{S_{maxj} - S_{minj}}; \forall j = 1, \dots, D_P] \quad (2)$$

Since large convolution kernels have larger receptive fields, they are used in the initial CNN layer to achieve better results. As for the problem that large convolution kernels tend to ignore local information, a classic strategy is adopted to solve it, i.e. as the network deepens, the convolution kernel is continuously reduced and the number of channels is increased, in order to find a balance in the extraction process of local information and global information. Since the physiological data of each channel do not contain much information, three-layer 1D-CNN and linear interpolation are finally selected to obtain physiological feature $F_P^0 \in R^{L \times D}$, maintaining simplicity while preventing overfitting caused by deep network.

B. Multi-modal Interaction Based on Dual Token

This module is designed to capture emotional cues and maintain global semantic consistency. The local information extractor with local token is designed for audio-visual modal interaction, aiming to extract the emotional information contained in the video content. The global information extractor with global token is designed for global semantic extraction, aiming to maintain global information consistency among the three modalities.

1) **Local Information Extractor:** The early interaction process of the visual and audio modal consists of N layers. At this stage, we can obtain the output $F_V^N, F_A^N, F_{Local}^N$ of the N -th layer from the inputs F_V^0, F_A^0 . Specifically, local token $F_{Local}^0 \in R^{l \times D}$ and global token $F_{Global}^0 \in R^{l \times D}$ are generated using random initialization. At the beginning of each layer, the local token is updated and regarded as a new modal. Then, new visual and auditory modal features are obtained by concatenating local token with visual and audio modal separately. After that, those three modalities are sent to their respective transformer blocks to obtain the output of this layer. This process will be repeated N times, where N is a hyperparameter. Taking the i -th layer as an example, the process is as follows:

$$F_{newL}^i = Update(F_{Local}^i, F_V^i, F_A^i) \quad (3)$$

$$F_{Local}^{i+1} = Transformer(F_{newL}^i) \quad (4)$$

$$F_V^{i+1} = Transformer(concat(F_{newL}^i, F_V^i)) \quad (5)$$

$$F_A^{i+1} = Transformer(concat(F_{newL}^i, F_A^i)), \quad (6)$$

where $i \in \{0, 1, \dots, N-1\}$. $Transformer(\cdot)$ denote the transformer block [54]. Inspired by the work of Mo et al. [55], the $update(\cdot)$ process uses the same architecture as the Fusion Block proposed in their work, as shown in Fig. 3. Since the number of features in audio/visual modal are numerous and the cost of interactions can become incalculable, modality-specific

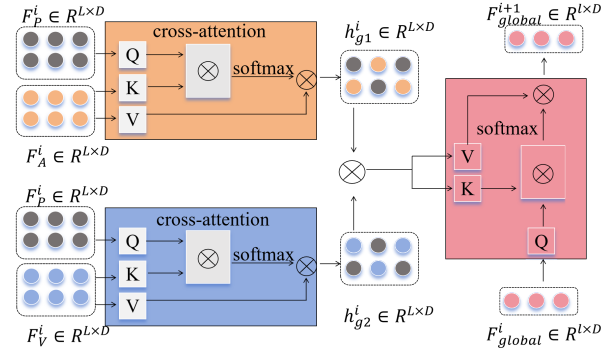


Fig. 4: An illustration of Storage Block.

aggregation tokens are used to summarize the information stored in audio/visual modal. We first initialize these tokens randomly and update them by cross-attention, obtaining new tokens $\overline{F_{V_i}} \in R^{l \times D}$ and $\overline{F_{A_i}} \in R^{l \times D}$. Then we use linear aggregation to conduct audio-visual interaction following Mo et al. [55]. Afterward, local token is updated by cross-attention. Formally, $update(\cdot)$ is represented as follows:

$$\overline{F_V^{i+1}} = CrossAttention(\overline{F_V^i}; F_V^i) \quad (7)$$

$$\overline{F_A^{i+1}} = CrossAttention(\overline{F_A^i}; F_A^i) \quad (8)$$

$$F_{VA}^i = W^V \overline{F_V^{i+1}} + W^A \overline{F_A^{i+1}} \quad (9)$$

$$F_{newL}^i = CrossAttention(F_{Local}^i; F_{VA}^i), \quad (10)$$

where W^V and W^A represent the learnable parameters. Through the local information extractor, the fine-grained correlation in the audio features and video features at the corresponding temporal segment is captured to extract the emotional information contained in the video content.

2) **Global Information Extractor:** This process also consists of N layers. At this stage, we can obtain the output F_P^N, F_{Global}^N of the N -th layer from the inputs F_V^0, F_A^0, F_P^0 . The physiological modal does not interact directly with the audio/visual modal at this time because the information in the audio/visual modality mostly represents common emotion, i.e. emotion of original video. Since this data is collected from each subject, the physiological modal can be considered to contain specific personal information. Early fusion of them may lead to information confusion, resulting in a decrease in model performance. However, performing multi-modal fusion directly at the end may cause insufficient fusion and inability to utilize multi-modal information fully. Therefore, we introduce global token to store the result of the interaction between physiological modal and audio/visual modal, and gradually update it at each layer. At the same time, with the continuous iteration of the visual-audio interact and update module, physiological modal is also continuously updated through the gated recurrent unit (GRU), gradually transforming low-level features into high-level features. As shown in Fig. 4, the process of the storage block is shown as follows:

$$h_{g1}^i = softmax(\frac{F_P^i (F_V^i)^T}{\sqrt{d}}) F_V^i \quad (11)$$

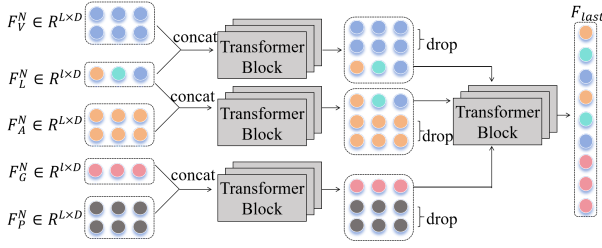


Fig. 5: An illustration of Hierarchical Fusion module.

$$h_{g2} = \text{softmax}\left(\frac{F_P^i (F_A^i)^T}{\sqrt{d}}\right) F_A^i \quad (12)$$

$$F_{Global}^{i+1} = \text{softmax}\left(\frac{F_{Global}^i (h_{g1} \times h_{g2})^T}{\sqrt{d}}\right) (h_{g1} \times h_{g2}) \quad (13)$$

where $i \in \{0, 1, \dots, N-1\}$, F_V^i, F_A^i, F_P^i represent visual, audio, and physiological features at the beginning of the i -th layer, F_{Global}^i is the global token of this layer, and $\times$ represents the vector outer product. First, the physiological features are used as queries, and let physiological features calculate cross-attention with visual and audio features respectively, obtaining h_{g1} and h_{g2} . Then, information in global token is stored using cross-attention. Furthermore, physiological feature are updated using GRU, without interaction with global token.

Motivated by the idea of bilinear aggregation, here we use outer product to capture the correlation between features. The outer product can capture the correlation between feature channels and simulate the interaction between modalities. [56] By calculating the outer product of h_{g1}^i and h_{g2}^i , the emotional information reflected in different dimensions at the same segment is captured and stored in F_{Global}^{i+1} . Unlike simple concatenation or addition, the outer product facilitates a fine-grained interaction across all feature dimensions. This means that the outer product response is enhanced only when the audio-visual modal exhibits high consistency across a particular emotional dimension. Therefore, by capturing these high-order correlations, F_{Global}^{i+1} effectively stores global semantic of a segment, which provides a stable reference for subject-independent task.

$$F_P^{i+1} = \text{GRU}(F_P^i) \quad (14)$$

C. Hierarchical Fusion Module

This module is used to gradually fuse cross-modal cues after controlled interaction, so that key emotion information can be retained, as shown in Fig. 5. Through this module, we can obtain F_{last} from $F_V^N, F_L^N, F_P^N, F_{Local}^N, F_{Global}^N$. First, the visual and audio features are concatenated with the local token to obtain the fused feature F_V^{fusion} and F_A^{fusion} . The physiological features are concatenated with the global token to obtain the fused features F_P^{fusion} . Then, $F_V^{fusion}, F_A^{fusion}, F_P^{fusion}$ pass through their respective transformers blocks to obtain the updated features:

$$F_V^{fusion} = \text{concat}(F_V'; F_{Local}') \quad (15)$$

$$F_A^{fusion} = \text{concat}(F_A'; F_{Local}') \quad (16)$$

$$F_P^{fusion} = \text{concat}(F_P'; F_{Global}') \quad (17)$$

$$F_M^{new} = \text{drop}(\text{Transformer}(F_M^{fusion})), M \in \{V, A, P\} \quad (18)$$

where $M \in \{V, A, P\}$ represents three modalities, and $\text{drop}(\cdot)$ means selective discarding. To more efficiently represent emotional information and keep crucial information, we utilize modal-special tokens to represent each modal and drop original features. Concatenating tokens and modal features aims to transfer important information from modal to tokens through the transformer, can remove redundant information irrelevant to emotion.

Unlike treating each modality equally, a hierarchical fusion method is used here. The local token has been used multiple times to update the audio/visual features, but the global token is only used at this stage to update physiological features. The information in the local token comes from the Local Information Extractor module. Local token has interacted with the audio/visual features before the hierarchical fusion module, and gradually updates itself. Further fusion will be performed here. The information in the global token comes from the Global Information Extractor module. Before the hierarchical fusion module, global token only records the process of multi-modal interaction, and does not update the physiological modal. The first fusion will be performed here. Through this hierarchical interaction method and drop mechanism, our model can gradually identify key emotional information and store it in tokens. Finally, $F_V^{new}, F_A^{new}, F_P^{new}$ are spliced and sent to transformer to obtain the final features used for emotion recognition, i.e. F_{last} .

$$F_{last} = \text{Transformer}(\text{concat}(F_V^{new}, F_A^{new}, F_P^{new})) \quad (19)$$

IV. EXPERIMENTS

A. Datasets

This paper conducts experiments on three emotion datasets with continuous annotation, namely CEAP [57], CASE [58] and AVDOS [59]. The scale, video type and physiological data type of these datasets are shown in TABLE I. The videos in the three datasets we selected all encode common emotional context that can systematically reduce inter-subject variability. CEAP is a continuous physiological and behavioral emotion annotation dataset for 360° VR videos. It contains 32 subjects and eight 60-second videos. CASE is a dataset of continuous affect annotations and physiological signals. It contains 30 subjects and 12 videos of 1-2 minutes each. AVDOS is an affective VR video dataset with multi-modal physiological signals and continuous self-ratings collected via a VR controller. It contains 37 subjects and thirty 30-second videos.

The division of datasets is consistent with the approach in CEAP, i.e. use Leave-One-Subject-Out Cross Validation (LOSOVC) under the experimental setup of subject-independent (SI). Based on previous research [11], we choose 2s as the length of a fine-grained segment and conduct experiments under this condition.

TABLE I: Details of three continuously-annotated emotion datasets.

Dataset	Scale&Video Type	Physiological data
CEAP	32 people 8 VR video clips of 60s	BVP/SKT/EDA HR/IBI
CASE	30 people 12 video clips of 1-2 minutes	BVP/SKT/ECG GSR/Respiration
AVDOS	37 people 30 VR video clips of 30s	EMG/PPG/IMU Skin contact (impedance)

B. Training Settings

The input shape of the image was fixed at 224×224, and the audio was resampled to 16kHz. The optimizer was set to Adam, with a learning rate of 0.001. Repeat times N was set to three. The learning rate was multiplied by 0.95 every two epochs. Training proceeded for 50 epochs with an early stopping strategy. The batch size was set to 30. The length l of local token and global token were 10, and the random seed was fixed at 42. All experiments were conducted on an NVIDIA GeForce RTX 4070 GPU 16GB.

C. Comparative Results

Comparisons with various existing methods were conducted across three datasets and performed detailed visualization analyses. As shown in Tables II, III, and IV, our model achieves superior performance under various conditions with the support of the dual token mechanism and hierarchical fusion module. Furthermore, the confusion matrix in Figure 6 and the heatmap analysis in Figure 7 reveal that the model performs better on high valence or high arousal classification, and no subject exhibits consistently abnormal results.

First, comparisons results with existing methods on three continuously-annotated emotion datasets are presented in TABLE II, with the best results highlighted in bold. Since there are few existing methods on these three datasets at this stage, we have added CNN [60], LSTM [61], and Transformer [54] for comparison. We conduct various classification experiments, it can be seen that with the support of dual token and hierarchical fusion module, the accuracy of our model has significantly improved.

Subsequently, in order to demonstrate our model's cross-subject generalization capability, we have conducted experiments under subject-independent condition, as show in Table III. Results show that our model can adapt to different types of videos and achieves improvements on all three datasets, especially in five-class classification. By incorporating stimulus information, our model can better learn how physiological signals change under a given video context. This structure reduces the model's uncertainty in cases where the correlation between the subjects' physiological responses and specific emotions is not obvious. Therefore, our model offers more significant improvements for complex tasks. In particular, the classification results of Arousal-3 are abnormally high on the CASE dataset. After analyzing the data distribution, we consider that this may be caused by the uneven distribution of the dataset. There are 44087 low arousal samples among

all 757200 samples, i.e. only 5.82% of the samples are low arousal, which leads to an overestimation of the model's results.

Finally, to demonstrate that our model remains competitive when compared against multi-modal methods, we reproduced ten recent multi-modal methods on three datasets and compared them with our method. As shown in TABLE IV, it is evident that our model outperforms all other methods across the CASE, CEAP, and AVDOS datasets. It can be seen that adding more modalities does not equate to performance improvement. Suboptimal integration method will fail to further improve model performance and may even lead to poorer performance. For example, MULT [46] uses three modalities, while SDT [48] uses only two modalities. However, SDT performs better on the CEAP dataset as illustrated in Table IV. Moreover, it is worth noting that Mood-Reader [34] and MDNet [35] are specifically designed for subject-independent recognition using only physiological multi-modal data, their performance varies greatly on different datasets. This phenomenon further demonstrates that combining audio-visual and physiological modal can achieve better and more stable performance, as our model has done.

In order to visually demonstrate the model's classification performance across different datasets, we plotted the confusion matrices in Figure 6. The results show that the model has better performance on high valence or high arousal classification. On CEAP dataset, though the classification results of each category are average, intensive/positive emotion classification still show higher accuracy. On AVDOS dataset, results on HVHA and LVHA show better performance, while results on others still leave room for improvement. On CASE dataset, results on each category are obviously different. HVLA and LVHA category achieve accuracies of 77% and 87%, while LVLH and Neutral are difficult to identify. Considering the scarcity of low emotion samples in this dataset, although this result is expected, it is still a shortcoming that deserves improvement.

In order to visually demonstrate the generalization ability of our method across different subjects, all the subject-independent classification results aggregated per subject of three datasets are presented in the form of heatmap in Fig. 7. As shown in Fig. 7, there is no situation where a user's classification results are always particularly good or particularly bad. It demonstrates that our model has a certain degree of universality among all subjects. Taking 5-class as an example, users with the worst results on the CASE, CEAP and AVDOS dataset are user 18, user 2, and user 28, with accuracy of 33.17%, 33.33%, 25.78% respectively. It implies that combining the common information contained in videos and the personal information can effectively alleviate the high inter-subject variability of physiological signals, thus improves the model's ability to recognize emotions for unknown subjects.

D. Ablation Experiment

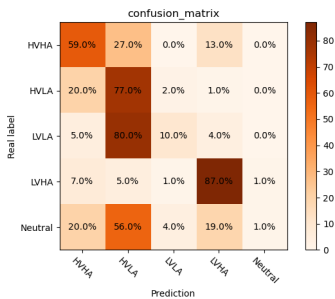
Ablation studies are conducted on three datasets using a subject-independent approach to investigate the efficacy of the proposed method. Specifically, backbone only utilizes

TABLE II: Comparison with existing results on three continuously-annotated emotion datasets under subject dependent experimental condition. '*' indicates that the result is taken from original paper.

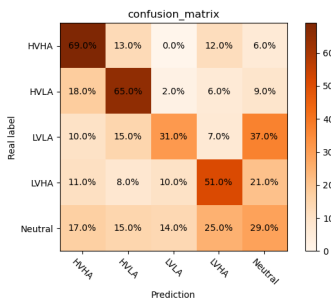
Dataset	Method	Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
		acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
CASE	CNN [60]	0.5356	0.6073	0.6841	0.7483	0.5672	0.6190	0.6563	0.6927	0.5301	0.6145
	LSTM [61]	0.4692	0.5847	0.5723	0.6861	0.3241	0.4209	0.2822	0.3390	0.3282	0.4540
	Transformer [54]	0.8199	0.8511	0.7627	0.8027	0.7472	0.7903	0.8136	0.8521	0.5752	0.6326
	Zhang [11]	*0.7701	*0.74	*0.8011	*0.79	*0.6183	*0.61	*0.6203	*0.61	0.5546	0.5211
	Zhang [12]	*0.7730	*0.722	*0.7772	*0.709	-	-	-	-	*0.5877	*0.482
	Ours	0.9261	0.9295	0.9024	0.9040	0.9031	0.9075	0.9168	0.9224	0.7931	0.8186
CEAP	CNN [60]	0.6846	0.5739	0.7137	0.6121	0.5117	0.3867	0.5685	0.4502	0.4049	0.2828
	LSTM [61]	0.6551	0.6031	0.7139	0.6560	0.5291	0.4755	0.5636	0.5002	0.4448	0.3735
	Transformer [54]	0.7414	0.7783	0.7624	0.8084	0.6190	0.6607	0.6379	0.6998	0.4931	0.5704
	Zhang [12]	*0.7186	*0.695	*0.7495	*0.729	-	-	-	-	*0.5693	*0.512
	Ours	0.9143	0.9140	0.9092	0.9109	0.8477	0.8555	0.8307	0.8407	0.7251	0.7537
	Ours	0.9143	0.9140	0.9092	0.9109	0.8477	0.8555	0.8307	0.8407	0.7251	0.7537
AVDOS	CNN [60]	0.4698	0.5305	0.4828	0.5405	0.4265	0.4665	0.4171	0.4466	0.2324	0.2779
	LSTM [61]	0.4705	0.5314	0.5380	0.6132	0.4668	0.5340	0.3315	0.3889	0.2462	0.2941
	Transformer [54]	0.5741	0.6229	0.7216	0.7670	0.4765	0.5345	0.5441	0.6115	0.3433	0.3919
	Ours	0.8212	0.8520	0.8619	0.8911	0.7641	0.7950	0.7690	0.8114	0.6724	0.7221
	Ours	0.8212	0.8520	0.8619	0.8911	0.7641	0.7950	0.7690	0.8114	0.6724	0.7221

TABLE III: Comparison with existing results on three continuously-annotated emotion datasets under subject independent experimental condition. '*' indicates that the result is taken from original paper.

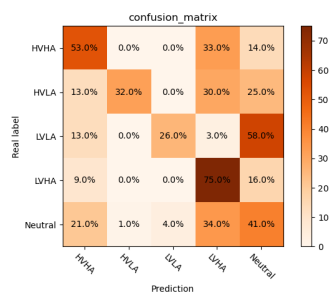
Dataset	Method	Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
		acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
CASE	CNN [60]	0.5249	0.5548	0.5242	0.5725	0.4457	0.4918	0.3839	0.4086	0.2805	0.2997
	LSTM [61]	0.5198	0.5630	0.5210	0.5731	0.3934	0.4334	0.3387	0.3661	0.2544	0.2974
	Transformer [54]	0.6835	0.7172	0.6273	0.6727	0.6204	0.6036	0.6955	0.7364	0.3201	0.3297
	Zhang [11]	*0.7637	*0.76	*0.7403	*0.72	*0.6015	*0.53	*0.6604	*0.65	0.4582	0.4449
	Wu [10]	*0.7857	*0.7774	*0.7498	*0.7310	*0.6664	*0.6485	*0.6632	*0.6178	0.4844	0.4681
	Dissanayake [62]	*0.7408	*0.7064	*0.7630	*0.7127	*0.6483	*0.6025	*0.6507	*0.6108	0.4545	0.4331
	Ours	0.8112	0.8412	0.7676	0.7965	0.6750	0.6712	0.8144	0.8293	0.5622	0.6078
CEAP	CNN [60]	0.6427	0.5828	0.6764	0.5808	0.4551	0.4191	0.4717	0.3923	0.2987	0.2598
	LSTM [61]	0.6500	0.6349	0.6630	0.5934	0.4462	0.4269	0.4379	0.4085	0.3012	0.2768
	Transformer [54]	0.6625	0.7199	0.6849	0.7291	0.4188	0.4133	0.5047	0.6054	0.3289	0.4311
	Quintero [63]	*0.7200	-	*0.6970	-	0.5674	0.5359	0.4976	0.4878	0.4061	0.3960
	Ours	0.8034	0.8173	0.7307	0.7378	0.6400	0.6729	0.5853	0.6359	0.4732	0.5213
AVDOS	CNN [60]	0.4349	0.4981	0.4695	0.5278	0.3978	0.4382	0.3172	0.3435	0.2420	0.2889
	LSTM [61]	0.4356	0.4981	0.5699	0.5790	0.3890	0.4436	0.4122	0.4873	0.2314	0.2787
	Transformer [54]	0.5315	0.5258	0.6304	0.6830	0.3551	0.3886	0.4512	0.5289	0.2820	0.3317
	Ours	0.7324	0.7754	0.7304	0.7831	0.5376	0.5808	0.5561	0.6201	0.4482	0.5019



(a) Confusion matrix of CASE



(b) Confusion matrix of CEAP



(c) Confusion matrix of AVDOS

Fig. 6: Confusion matrix visualization results for three different datasets using proposed method DTHFormer, for the subject-independent prediction of 5-class

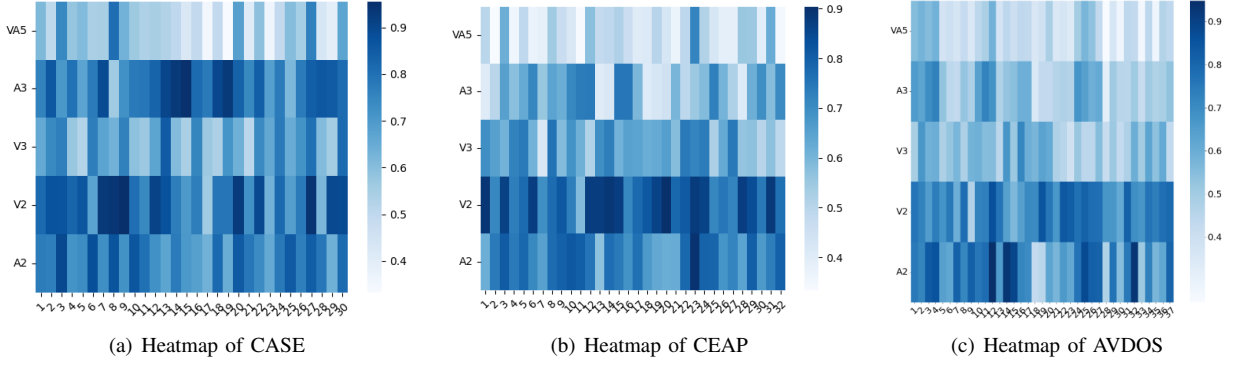


Fig. 7: Heatmap depicting all classification results aggregated per subject of three datasets.

TABLE IV: Comparison results of 5-class with multi-modal methods under subject independent condition

Method	CASE		CEAP		AVDOS	
	acc	w-f1	acc	w-f1	acc	w-f1
MULT [46]	0.5428	0.5892	0.4547	0.4966	0.4164	0.4665
Huang [43]	0.5161	0.5610	0.4576	0.4944	0.4112	0.4615
VAANET [38]	0.5281	0.5698	0.4257	0.4662	0.4278	0.4783
VA-Transformer [18]	0.5318	0.5764	0.4588	0.5046	0.4089	0.4578
SAMS [47]	0.5577	0.5977	0.4644	0.5093	0.3189	0.3544
SDT [48]	0.5304	0.5641	0.4664	0.5155	0.3652	0.3839
Mood-Reader [34]	0.5595	0.6049	0.4706	0.4740	0.2744	0.3242
MDNet [35]	0.5501	0.4774	0.3297	0.4010	0.3485	0.3894
Salas-Cáceres [49]	0.4844	0.5212	0.3601	0.4562	0.3483	0.4009
MultiDepNet [50]	0.5441	0.5933	0.4522	0.4617	0.3773	0.4151
ours	0.5622	0.6078	0.4732	0.5213	0.4482	0.5019

multi-modal feature extractor. For this part, Local Information Extractor is denoted as Local, Global Information Extractor is denoted as Global, the Hierarchical Fusion module is denoted as HF. The other experimental settings for the ablation experiments were consistent with those in the previous experiments.

The ablation experimental results are shown in TABLE V, TABLE VI and TABLE VII, proving the proposed method's effectiveness. Compared with binary classification, improvements on 5-class classification are more significant. Moreover, directly using the extracted information of the three modalities gives the worst results under any conditions. It demonstrates that the proposed method DTHFormer is better at performing complex classification tasks since we mitigates information confusion. Each module has achieved certain results in different classification tasks, while in some cases double modules lead to worse results compared with single module. The reason may be that the lack of module causes the model to pay less attention to the information of certain modal, which may be important. As for 5-class classification on the CEAP dataset, we consider that the information contained in 360° videos is redundant to a certain extent, which leads to a worse

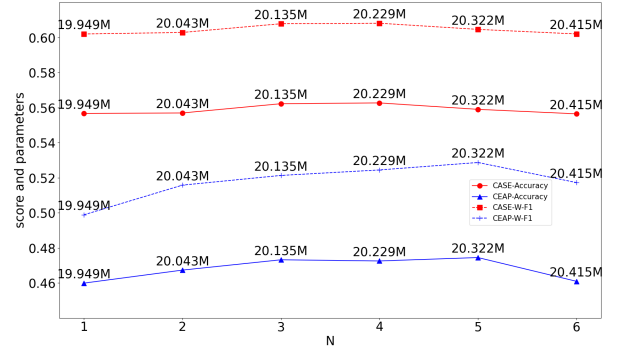


Fig. 8: Effect of repeat times N on different datasets.

classification result without adding global feature extractor.

In order to explore the effect of multi-modal approach compared with unimodal approach, we conducted classification experiments on visual, audio and physiological modal respectively, and compared the results with the multi-modal approach. As shown in TABLE VIII, results of multi-modal fine-grained emotion recognition are better than the results of unimodal on three datasets. Multi-modal emotion recognition can combine the information of each modal and analyze the temporal information to better extract the emotional clues in each modal, thereby more accurately recognizing complex emotions.

To explore the effect of Hierarchical Fusion module, we replaced this module with some commonly used fusion modules, i.e. early fusion, late fusion and hybrid fusion. As shown in TABLE IX, Hierarchical Fusion module achieved better results among various experiments on three datasets.

A Further analysis is performed on repeat times N in the multi-modal interaction module. Fig. 8 illustrates the changing curves of accuracy and weighted-F1 as N changes. All experiments with N equal to 1-6 can be completed on an NVIDIA GeForce RTX 4070 GPU. For CASE dataset, the model performance is optimal when N = 4 and suboptimal when N = 3. For CEAP dataset, the model performance is optimal when N = 5 and suboptimal when N = 3. As N continues to increase, the performance on both datasets declines. Moreover, when N equals 1, the number of parameters in the model is 19,949,575, when N equals 6 the number of parameters in the

TABLE V: Ablation Experiment on CASE dataset

Extractor	Module			Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
	Global	Local	HF	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
✓				0.7947	0.8225	0.7389	0.7437	0.6330	0.6449	0.8060	0.8211	0.5354	0.5840
✓	✓			0.8098	0.8360	0.7587	0.7758	0.6604	0.6586	0.8069	0.8239	0.5530	0.6013
✓		✓		0.8073	0.8320	0.7568	0.7727	0.6602	0.6624	0.8080	0.8245	0.5542	0.5973
✓			✓	0.8104	0.8351	0.7548	0.7719	0.6580	0.6583	0.8015	0.8214	0.5580	0.6044
✓	✓		✓	0.8066	0.8335	0.7646	0.7931	0.6658	0.6657	0.8020	0.8178	0.5456	0.5964
✓		✓	✓	0.8096	0.8373	0.7660	0.7906	0.6680	0.6658	0.8116	0.8225	0.5528	0.6077
✓	✓	✓		0.8088	0.8332	0.7638	0.7897	0.6696	0.6684	0.8134	0.8286	0.5431	0.5918
✓	✓	✓	✓	0.8112	0.8412	0.7676	0.7965	0.6750	0.6712	0.8144	0.8293	0.5622	0.6078

TABLE VI: Ablation Experiment on CEAP dataset

Extractor	Module			Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
	Global	Local	HF	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
✓				0.7930	0.8096	0.7161	0.7238	0.6226	0.6586	0.5725	0.6189	0.4535	0.4977
✓	✓			0.8028	0.8131	0.7255	0.7305	0.6316	0.6655	0.5762	0.6172	0.4680	0.5133
✓		✓		0.8014	0.8120	0.7220	0.7274	0.6270	0.6579	0.5798	0.6181	0.4651	0.5103
✓			✓	0.8028	0.8185	0.7178	0.7260	0.6228	0.6455	0.5755	0.6212	0.4579	0.5058
✓	✓		✓	0.8026	0.8127	0.7271	0.7341	0.6376	0.6646	0.5808	0.6258	0.4818	0.5231
✓		✓	✓	0.8031	0.8163	0.7240	0.7346	0.6286	0.6612	0.5789	0.6235	0.4599	0.5135
✓	✓	✓		0.7995	0.8139	0.7155	0.7323	0.6359	0.6704	0.5846	0.6264	0.4586	0.5046
✓	✓	✓	✓	0.8034	0.8173	0.7307	0.7378	0.6400	0.6729	0.5853	0.6359	0.4732	0.5213

TABLE VII: Ablation Experiment on AVDOS dataset

Extractor	Module			Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
	Global	Local	HF	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
✓				0.7036	0.7362	0.7128	0.7648	0.4928	0.5302	0.5318	0.5953	0.4139	0.4587
✓	✓			0.7216	0.7558	0.7206	0.7732	0.5216	0.5638	0.5382	0.6018	0.4361	0.4902
✓		✓		0.7299	0.7626	0.7243	0.7712	0.5102	0.5539	0.5437	0.6124	0.4266	0.4776
✓			✓	0.7122	0.7479	0.7217	0.7723	0.4915	0.5329	0.5343	0.6063	0.4312	0.4862
✓	✓		✓	0.7289	0.7628	0.7203	0.7729	0.4979	0.5583	0.5445	0.6135	0.4248	0.4796
✓		✓	✓	0.7171	0.7548	0.7226	0.7719	0.5073	0.5605	0.5381	0.6116	0.4347	0.4926
✓	✓	✓		0.7261	0.7617	0.7242	0.7756	0.5258	0.5617	0.5422	0.6056	0.4259	0.4719
✓	✓	✓	✓	0.7324	0.7754	0.7304	0.7831	0.5376	0.5808	0.5561	0.6201	0.4482	0.5019

TABLE VIII: Comparison of the effects of unimodal and multi-modal

Dataset	Modal	Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
		acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
CASE	Visual	0.7121	0.7475	0.6450	0.6973	0.6170	0.6706	0.6956	0.7408	0.3805	0.4211
	Audio	0.6168	0.6248	0.6317	0.6542	0.5075	0.5125	0.5521	0.5553	0.3197	0.3069
	Physiological	0.6835	0.7172	0.6273	0.6726	0.6208	0.6703	0.6955	0.7364	0.3465	0.3518
	All	0.8112	0.8412	0.7676	0.7965	0.6750	0.6712	0.8144	0.8293	0.5622	0.6078
CEAP	Visual	0.6996	0.7006	0.6859	0.7268	0.4832	0.5059	0.5004	0.5325	0.3293	0.3518
	Audio	0.6559	0.6683	0.6515	0.6782	0.4554	0.4666	0.4522	0.4933	0.2477	0.2920
	Physiological	0.6658	0.7232	0.6865	0.7100	0.4240	0.4165	0.5049	0.6048	0.3286	0.4307
	All	0.8034	0.8137	0.7307	0.7378	0.6400	0.6729	0.5853	0.6359	0.4732	0.5213
AVDOS	Visual	0.6168	0.6504	0.6233	0.6487	0.4674	0.5057	0.4586	0.5281	0.3408	0.4012
	Audio	0.6125	0.5834	0.6304	0.6831	0.4317	0.4457	0.4146	0.4718	0.3053	0.3242
	Physiological	0.5601	0.6110	0.5938	0.6413	0.4013	0.4436	0.4124	0.3989	0.2670	0.2679
	All	0.7324	0.7754	0.7304	0.7831	0.5376	0.5808	0.5561	0.6201	0.4482	0.5019

TABLE IX: Comparison of using different fusion method

Dataset	Method	Valence-2		Arousal-2		Valence-3		Arousal-3		5-class	
		acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1	acc	w-f1
CASE	Late	0.8102	0.8374	0.7647	0.7816	0.6319	0.6563	0.7937	0.8095	0.5432	0.5880
	Early	0.8092	0.8374	0.7626	0.7933	0.6673	0.6621	0.8048	0.8215	0.5549	0.6046
	Hybrid	0.8091	0.8373	0.7611	0.7782	0.6574	0.6599	0.8091	0.8220	0.5548	0.5996
	Hierarchy	0.8112	0.8412	0.7676	0.7965	0.6750	0.6712	0.8144	0.8293	0.5622	0.6078
CEAP	Late	0.7888	0.8025	0.7154	0.7324	0.6014	0.6387	0.5490	0.5932	0.3898	0.4606
	Early	0.8004	0.8048	0.7209	0.7292	0.6302	0.6726	0.5820	0.6267	0.4717	0.5113
	Hybrid	0.7891	0.8038	0.7179	0.7271	0.6015	0.6354	0.5638	0.6156	0.4069	0.4684
	Hierarchy	0.8034	0.8137	0.7307	0.7378	0.6400	0.6729	0.5853	0.6359	0.4732	0.5213
AVDOS	Late	0.7212	0.7573	0.7196	0.7713	0.5203	0.5706	0.5077	0.5872	0.3492	0.4075
	Early	0.7246	0.7621	0.7236	0.7745	0.5268	0.5796	0.5123	0.5893	0.4344	0.4818
	Hybrid	0.7258	0.7654	0.7254	0.7736	0.5214	0.5728	0.5112	0.5887	0.3728	0.4291
	Hierarchy	0.7324	0.7754	0.7304	0.7831	0.5376	0.5808	0.5561	0.6201	0.4482	0.5019

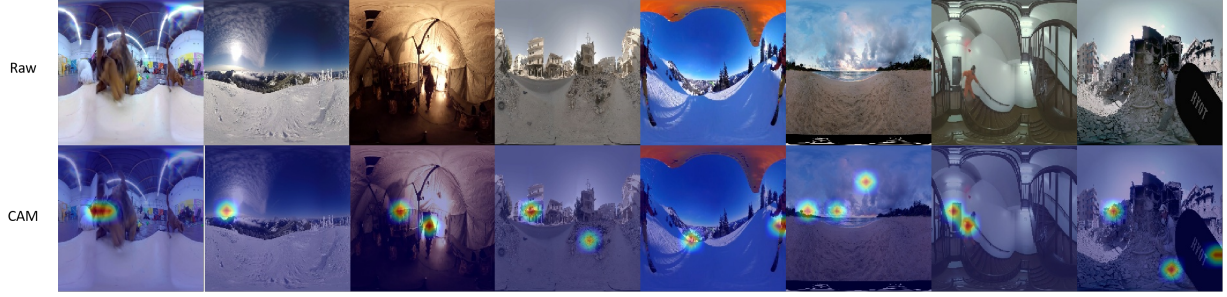


Fig. 9: Visualization results on dataset CEAP.

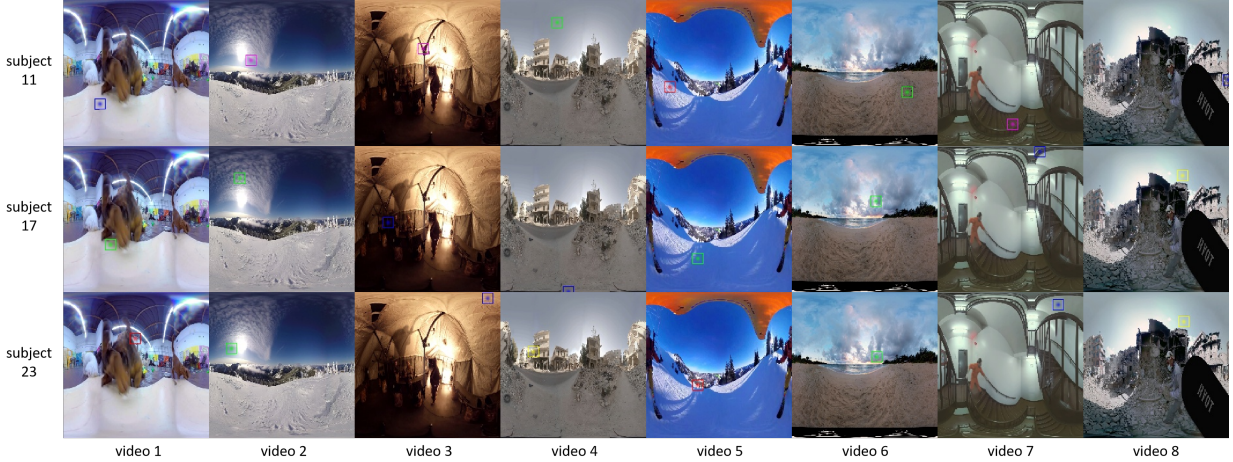


Fig. 10: Visualization of the gaze points and corresponding emotions of different subjects, taking subjects 11, 17, and 23 as examples. The location of the gaze point is highlighted with a colored box, and the color of the gaze point represents the subject's emotion at the current moment. Red represents HVHA, green represents HVLA, yellow represents LVLA, blue represents LVHA, and magenta represents neutral.

model is 20,415,325. Meanwhile, when N is greater than 3, the cost of increasing training time is greater than the benefits gained from increasing the parameter N . Considering the cost of time and computing resources, we chose $N = 3$ at last.

V. VISUALIZATION AND DISCUSSION

In order to further explore the effectiveness of the proposed model with dual token and hierarchical fusion, we utilize class activation mapping to visually highlight the regions of interest identified by our model. The experiments were conducted on dataset CEAP, as shown in Fig. 9. The results in the figure

show that the proposed model has a strong ability to perceive semantic objects in the video. Besides, in most cases, the proposed model can focus on multiple core areas that may evoke emotions. For example, in the fifth video, the model focuses on the characters in the picture and the snow track directly in front of the line of sight; in the sixth video, the model focuses on the coastline and the changing clouds.

Furthermore, to further analyze the relationship between key video regions and the emotions of different subjects, we visualized the subjects' gaze points and their emotions at the corresponding moments. The location of the gaze point is

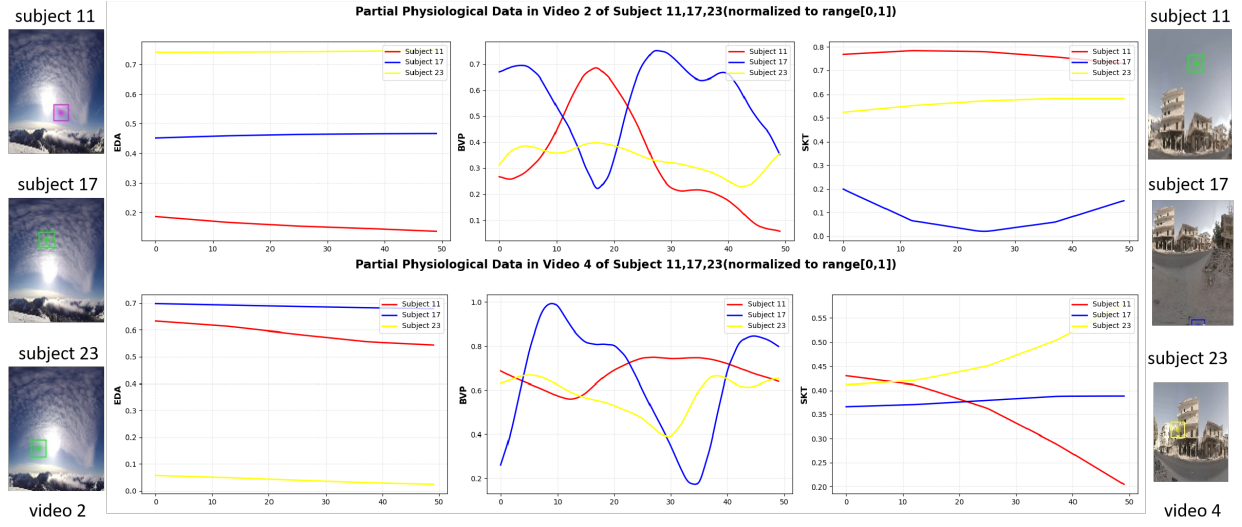


Fig. 11: Partial physiological data of subject 11,17,23, when they were watching video 2 and video 4. All selected data were normalized to range [0,1], following the processing method of the CEAP dataset authors.

highlighted by a colored box, and the color of the gaze point represents the subject's emotion at that moment, as shown in Fig. 10. We can observe that the emotions exhibited by different subjects may differ, and such difference may possibly be attributed to differences in their gaze position(video 4). On the other hand, even though the gaze area is roughly the same, different subjects may still express different emotions(video 2).

In response to the two phenomena mentioned above, we analyzed the physiological data of three subjects in the same segment, as shown in Fig. 11. In video 2, all three subjects gaze near the sun, but the emotion expressed by subject 11 is different from the other two. From physiological perspective, subject 11 maintained a stable high SKT and low EDA, suggesting a possible comfortable state(Neutral). Subject 17 exhibited dramatic changes in BVP and a slight increase in EDA, indicating that his physiological system was actively responding to the video content(high valence). In this case, the characteristics of the subjects themselves may play a more critical role. That is, when a specific modal exhibits synchronized patterns across different subjects, its information density may decrease due to redundancy. Our proposed hierarchical fusion module can gradually identify and preserve key emotional information through differentiated multi-modal interaction forms.

In Video 4, although the overall emotional tone of Video 4 is LVLA (as reported in CEAP [57]), the three subjects still express three different emotions while their electrodermal activity (EDA) are all stable. It can be seen that similar physiological responses may paired with different emotional labels. Subject 23's gaze coincides with the core region highlighted by our model in Fig. 9, and the subject's emotion is LVLA, aligning with the overall emotional tone of Video 4. In this case, the emotional semantic information provided by the audio-visual modal may play an important role. Through the dual-token structure, our model is able to better learn the common emotions expressed by different subjects.

Overall, the emotional information contained in the audio-

visual modal may differ from the subject's own feelings. Directly integrate audio-visual data and physiological data may cause information confusion, which may lead to misjudgment. Through dual token interaction and hierarchical fusion module, our model is able to capture both common emotions and individual differences, providing more robust and accurate emotion recognition results.

VI. CONCLUSION

This paper proposes a novel approach (DTHFormer) for fine-grained emotion recognition. The approach aims at integrating information in emotional stimulus with physiological information by conducting indirect multi-modal interaction and hierarchical fusion. It starts with indirect multi-modal interaction based on dual token. Specifically, utilizing local token to assist the interaction of modalities with high similarity. At the same time, utilizing global token to assist the interaction of modalities with low similarity. Then conduct hierarchical fusion to preserve key emotional information across multi-modal data, enabling the model to better understand emotional information. The proposed method is tested on three emotion datasets with continuous annotation across various classification experiments. In addition to improving the accuracy and stability of fine-grained emotion recognition, our method also provides a possible application approach in wearable device applications. Specifically, this capability extends to scenarios with limited data availability, where fine-grained multi-modal emotion modeling can be realized by integrating short-term observations with external stimulus cues, without relying on long-term subject-specific baselines.

This paper focuses on standard videos and VR videos as stimulus, however, actually it can be extended to other types of stimuli, like image, audio, and text. When dealing with different types of stimuli, it may be necessary to use a suitable feature extractor and perform special processing on the missing modal. Furthermore, further exploration is needed in modeling the temporal dynamic evolution of emotions. The

temporal causal modeling mechanism may be a good choice to investigate this issue in our future work.

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